

Comparison of Methods to Estimate Missing Values in Monthly Rainfall Data

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Abstract - Missing data is common in data analytics, including rainfall, and can reduce accuracy through biased estimates that lead to invalid conclusions. Many methods have been proposed to handle missing data but the same method might not be suitable for every dataset. This challenge was taken up here as the presentation of a comparative study of popular methods to handle the issue of missing values. Popular methods of handling missing data values include removal of instances with missing values, arithmetic mean (AM), normal ratio (NR), nearest neighbor (NN) and inverse distance (ID). We proposed a method of estimating missing values called *step-5 simple moving average (step-5 SMA)*. This applies simple moving average (SMA) principles with consideration of the El Niño-Southern Oscillation (ENSO) phenomenon. After enhancing the training set, our method was used to model monthly rainfall forecasts utilizing two supervised machine learning algorithms as support vector machines (SVM) and *k*-nearest neighbor (*k*-NN). We used monthly precipitation data (in mm) gathered between 1953 and 2013 from 26 water measurement stations of the Meteorological Department located in Northeast Thailand. After evaluating by MAE and RMSE, results showed that monthly rainfall forecasters developed by the training set that removed observations with missing values returned the lowest performance. Enhancing the quality of the training set using our *step-5 SMA* gave a better performance than the other missing value estimation methods.

Keywords— *Missing values, rainfall data, supervised machine learning, rainfall forecasting models, MAE, RMSE*

I. INTRODUCTION

Rainfall is an essential source of fresh water for all living creatures to survive. Historical rainfall data can be used to develop hydrological agricultural models to improve the management of water storage areas and provide advanced warning of flash flooding [1-3].

In general, historical datasets of rainfall have been used as a main source data for modeling of future rainfall volumes forecaster [4-5]. However, one common issue in real-world rainfall datasets is missing values as a result of various reasons, including human error during data entry or instrument failures during data storage [6-8]. Missing data reduce the accuracy and precision of predictive models and can lead to biased estimates and invalid conclusions. Therefore, historical rainfall datasets require improvement by adding standardized values to fill missing entries before

modeling rainfall volumes forecaster [9-13]. Many ‘filling’ methods have been proposed for missing data [9-13]. Ideally, the filling method chosen might be suitable for each particular dataset [14]. Some filling methods are inherently flawed and introduce more bias than they remove [15-16]. Currently, no method can completely handle missing data in every dataset. Research into this complex issue is continuing, including rainfall forecasting. Consequently, handling of missing value in historical rainfall data is also a challenge for this study.

In this study, we compared five standard filling methods for rainfall datasets as removal of instances with missing data [14], arithmetic average (AA) [7, 17-18], normal ratio (NR) [7, 17-18], nearest neighbor (NN) [17-18], and inverse distance (ID) [17-18]. These methods were also compared to our proposed method based on simple moving average (SMA) principle to evaluate their efficiencies in rainfall forecasting models. Two supervised machine learning algorithms as support vector machines (SVM) and *k*-nearest neighbor (*k*-NN) were applied. Mean absolute error (MAE) and root mean square error (RMSE) were used to evaluate rainfall forecasts.

The remainder of this study is organized as follows. Section 2 is the related work. Section 3 describes dataset and section 4 is the proposed method. Section 5 details the experimental setup. Section 6 showed experimental results and discussion, while the conclusion is in section 7.

II. RELATED WORK

Missing data are a serious issue in hydrological modeling, particularly in rainfall forecasting because the lack of information reduces forecasting power and may lead to selection bias. Many solutions of ‘filling’ missing values have been proposed to resolve this issue. Some studies could be described as follows.

Nkuna and Odiyo [19] proposed a method of filling missing rainfall data using an artificial neural network (ANN) with RBF kernel function. They used shuffled complex evolution (SCE) to identify the ANN’s best settings, and root mean square error (RMSE) was utilized as an objective function in the phase of calibration. Results of their prediction

analyzed by the Nash-Sutcliffe efficiency (NSE) ranged from 0.55 to 0.85, with percentage bias -2% to 23%. In the validation process, the NSE ranged between 0.49 and 0.75 with percentage bias 2% to 19%. They concluded that an ANN might be useful for estimating missing rainfall figures in their dataset.

Sanusi et al. [20] compared the performance of four methods to estimate missing values in monthly rainfall data as normal ratio, modified normal ratio, arithmetic mean, and correlation coefficient weighting. The correlation coefficient, the root mean squared error, the mean absolute error, and the S-index are then used to compare the performance of various methods. Results showed that the modified normal ratio was effective in estimating missing precipitation data.

Le et al. [21] mentioned that lack of good quality rainfall data can cause to reduce analysis efficiency and lead to biased results. Errors in estimating missing records due to faulty measurement processes of rainfall at neighboring stations cannot be ignored. Spatial interpolation is a common technique that is widely used for filling gaps in daily rainfall series by estimating the unknown rainfall from known data recorded by adjacent stations. They summarized the existing techniques for filling data including inverse distance, normal ratio, modified normal ratio, arithmetic mean, coefficient of correlation weighting, inverse exponential weighting, aerial precipitation ratio, kriging, thin-plate-spline, multiple linear regression, expectation maximization, parametric ordinary least-squares regression, non-parametric ranked regression and geometric mean functional regression.

Many methods for missing data handling have been proposed; however, several of these solutions are inherently flawed and may introduce more bias than they remove. Currently, no method can completely handle missing data in every dataset. Research into this complex issue is continuing, including rainfall forecasting.

III. DATASET

This study used monthly precipitation data (in mm) gathered between 1953 and 2013 from 26 water measurement stations of the Meteorological Department located in Northeast Thailand. The dataset consisted of 21 variables as 11,160 records. The meaning of each variable is explained in Table 1.

IV. METHODS OF ESTIMATING MISSING RAINFALL DATA

Several methods are applied to estimate missing rainfall data. In this study, some standard methods are also compared with our proposed method. These standard methods include removal of instances with missing data, arithmetic mean (AM), normal ratio (NR), nearest neighbor (NN) and inverse distance (ID), while our proposed method applies the simple

TABLE I. DETAILS OF DATSET

Variable	Meaning
Station	Water metering station code
Area_RiverWatershed	Three regions as Mekong River watershed, Chi River watershed and Mun River watershed
Area_Zone	Topography
Year	Year of storage
Month	Month of storage
Southern Oscillation Index (SOI)	A standardized index based on the observed sea level pressure differences between Tahiti and Darwin, Australia
Oceanic Nino Index (ONI)	A measure of the departure from normal sea surface temperature in the East-central Pacific Ocean as an indicator of El Niño and La Niña
Multi ENSO Index (MEI)	A combination of oceanic and atmospheric variables facilitated in a single index as an assessment of ENSO
Sea Surface Temperature (SST)	Water temperature near the ocean surface
Western North Pacific Monsoon Index (WNPPI)	The Pacific counterpart to the SAM, identified by strong rainfall over the subtropical Western North Pacific Ocean
Indian Summer Monsoon Index (IMI)	Rainfall over the subtropical Indian Ocean
Pressure	Value of barometric pressure
Relative	Value of relative humidity
Wind	Strength of wind
Temperature	Temperature
Wind_direction	Direction of wind
PhaseMEI	Index value of MEI
PhaseONI	Index value of ONI
Rain	Rain volume
dRain	Monthly rainfall
Rain_Mean	Average rainfall

moving average (SMA) principle [22], Meanings of the Water Measurement Data Variables with consideration of the El Nino-Southern Oscillation (ENSO) phenomenon [23].

A. Removal of Instances with Missing Data

In this method, the missing observations are ignored, and analysis is only done on the variables present.

B. The Arithmetic Mean (AM)

The AM method [17-18] is to calculate the arithmetic mean value of adjacent stations, and then this value is used to fill the missing value. The AM equation can be given by:

$$p_i = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

where x_i is the observed data at the i -th adjacent station, while n is the total number of adjacent stations. Finally, we can get p_i that refers to the estimated value of the missing value at t target station.

C. The Normal Ratio (NR)

The NR method [17-18] was introduced by Paulhus and Kohler in 1952. Later, it was modified by Young in 1992. The NR concept is the mean ratio of data between a target station and the i -th adjacent station. If any adjacent stations have

normal yearly rainfall and stream flow data 10% higher than the considered station, this method is often applied. The NR equation for estimating the missing value can be given by:

$$p_i = \frac{1}{n} \sum_{i=1}^n \frac{R_t}{R_i} x_i \quad (2)$$

where n is the total number of adjacent stations. R_t is to the yearly rainfall at the target station, R_i is the yearly rainfall at the i -th adjacent station.

D. The Nearest Neighbor (NN)

The NN [17-18] is also used for estimating and filling missing values, and involves selecting a nearest neighboring station and using its observations to fill missing values in the local station. It has confirmed that it is of the simplest approaches. A nearest neighbor station can be selected by considering the station with the highest correlation to the target location. As neighbors are computed by similarity analysis, the features should be natural number. After obtaining the value from a nearest neighboring station, this value can be used directly without making any change. The equation used for calculating the similarity between two or more data vectors as cosine similarity can be given by:

$$Sim(A, B) = \frac{\sum_{i=1}^n A_i \times B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}} \quad (3)$$

where A is the considered instance record and B is the compared instance record. A and B consist of many features; therefore, these features are computed to give similarity scores of A and B . The degree of similarity should be between -1 and 1, where -1 denotes perfect dissimilarity and 1 denotes perfect similarity.

E. The Inverse Distance (ID)

The ID method [17-18] is most commonly used to estimate the value of missing data. The ID concept is based on the distance between the target station and adjacent stations. Nearer stations show better correlation with the target station compared to those further away. The equation for estimating the missing value can be given by:

$$p_i = \frac{\sum_{i=1}^n \frac{x_i}{d_{i,t}}}{\sum_{i=1}^n \frac{1}{d_{i,t}}} \quad (4)$$

where the distance between the target station and the i -th adjacent station is denoted as $d_{i,t}$.

F. The Propose Method: Step-5 Simple Moving Average (Step-5 SMA)

Our proposed method is based on the moving average principle with consideration of the ENSO phenomenon, where the ENSO phenomenon affects global climate, including the occurrence of rainfall as a recurring pattern that generally changes every five years. The transition period of the ENSO phenomenon is taken into consideration for our five-year moving average method. The equation for SMA can be given by:

$$SMA = \frac{AV_1 + AV_2 + \dots + AV_n}{n} \quad (5)$$

where AV_n is the average value of factor that is considering at period n , and n is the number of total periods. In this study, n should be 5. Therefore, our proposed method is called *step-5 simple moving average (Step-5 SMA)*.

Suppose the SST value of some months in 2013 is missing. We will use the average values of the SST between 2008 and 2012 for estimating the SST value of those months in 2013. An example of the use of our proposed method is shown in Table 2.

TABLE II. EXAMPLES OF ESTIMATING DATA VALUE BY THE STEP-5 SMA METHOD

Year	Average Value of SST Value	Adding SST Value
2008	24.4	24.4
2009	24.9	24.9
2010	25.8	25.8
2011	25.9	25.9
2012	26.2	26.2
2013	?	$\frac{24.4 + 24.9 + 25.8 + 25.9 + 26.2}{5} = 25.44$

V. EXPERIMENTAL SETTING

After applying the methods of estimating missing values mentioned above, rainfall data with missing values added were used for rainfall forecast modeling. Two supervised learning algorithms were applied as k -NN and SVM.

To obtain the most appropriate models of rainfall forecasting and also explain the study results clearly, a 10-fold cross validation was applied. This approach was applied to appraise predictive models by separating the original sample into 10 subsamples. Nine subsamples are used as a training set to train the model, and the rest subsample is used as a validation set to appraise the model. The overview of experimental setting method can be presented as

A. k -NN

This algorithm is a lazy learner because its learning is driven on analogy. When the sample for which neighbors are to be computed is available, k -NN can be employed [24].

k -NN is computed for a given instance t for which the class label is to be predicted. The features are preferred to be natural number as neighbors are computed using the distance metric.

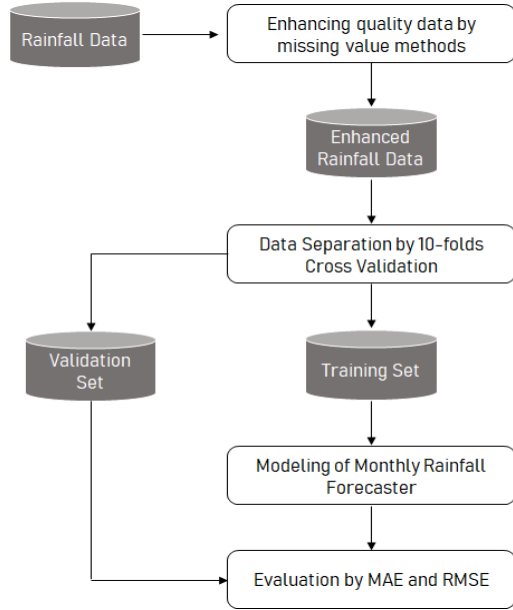


Fig. 1. The experimental setting method

To choose the value of k , it depends on the number of samples (N) in the training dataset. In general, k can be estimated by $\sqrt{\text{N}}$. The Euclidean Distance (ED) is also applied to calculate the distance between two points, and can be given by:

$$d(X, Y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (6)$$

where x_i is a data point from the training dataset and y_i is a data point from the test dataset for which the class is to be predicted. The class to which the maximum neighbors belong is considered for the input data.

B. SVM

SVM introduced for solving regression problems is called *support vector regression (SVR)* [25-26]. The SVR output is a real number with unlimited possibilities. Therefore, a tolerance limit is set for the SVM to minimize $\|w\|^2$ and reduce the model complexity. Here, the slack variables ξ_i and ξ'_i were applied to calculate the difference of training data outside the sensitive zone, where $i = 1, 2, \dots, n$.

Minimize:

$$\frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\xi_i + \xi'_i) \quad (7)$$

Subject to:

$$\begin{cases} y_i - f(x_{i,w}) \leq \varepsilon + \xi'_i \\ f(x_{i,w}) - y_i \leq \varepsilon + \xi_i \\ \xi_i, \xi'_i \geq 0, \dots, n \end{cases} \quad (8)$$

This study applied two kernel functions for the SVR model as RBF and sigmoid, represented as follows:

$$\text{RBF: } \exp(-\gamma \|X_i - X_j\|^2) \quad (9)$$

$$\text{Sigmoid: } \tanh(\gamma X_i \times X_j) + C \quad (10)$$

where $K(X_i, X_j) = \phi(X_i, X_j)$ and γ is the adjustable parameter.

VI. EVALUATION AND DISCUSSION

This study applied two metrics as MAE [27] and RMSE [27] to evaluate the performance of monthly rainfall forecasts. The training set used to develop the model improved the quality of missing values.

MAE [27] is a widely used evaluation formula that measures how far predicted values differ from observed values. The equation for MAE is given by:

$$MAE = \frac{1}{n} \sum_{i=1}^m |a_i - x_i| \quad (11)$$

where x_i is the predicted value and a_i is the actual value.

RMSE [27] is a quadratic scoring mechanism that calculates the average magnitude of the error. RMSE considers the prediction error by measuring the distance of data points from the regression line. RMSE is commonly used in regression analysis (e.g. rainfall forecasting) to verify experimental results. The equation for RMSE is given by:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (a_i - x_i)^2} \quad (12)$$

where x_i is the predicted value and a_i is the actual value.

Both MAE and RMSE could be ranged from 0 to ∞ as “negatively oriented” scores. Lower values give more accurate results [27].

The results of forecasting monthly rainfall that the models were developed by using the enhanced training set by removing instances of missing values using the AM, NR, NN, ID and step-5 SMA methods can be shown in Table 3. The overview of comparing results of methods to estimate missing values in monthly rainfall data through considering performance of rainfall forecasting models based on MAE and RMSE can be presented as Fig.2 and Fig. 3.

The k -NN, SVR with RFB, and SVR with sigmoid models of monthly rainfall forecasting developed by using the training set that removed observations with missing values returned low performance. This is because removing

observations with missing values introduces bias in the forecasting model. Improving the quality of the training dataset by adding appropriate values reduces bias. Therefore, monthly rainfall forecasts developed using our enhanced quality training set with added appropriate values returned better results compared to the training set that removed observations with missing values, with cumulative totals of MAE at 50.05% and RMSE at 56.91%.

TABLE III. THE EXPERIMENTAL RESULTS

Monthly Rainfall Forecasters	Methods for Handling Missing Value	MAE Score	RMSE Score
<i>k</i> -NN	Removing Instances	18.36	20.63
	AM	16.56	18.55
	NR	15.56	17.55
	NN	14.76	16.60
	ID	14.76	16.70
	Step-5 SMA	14.56	16.60
SVM with RBF	Removing Instances	14.991	16.78
	AM	9.70	10.07
	NR	9.70	10.07
	NN	7.51	8.16
	ID	7.51	8.01
	Step-5 SMA	7.41	7.86
SVM with Sigmoid	Removing Instances	15.59	17.78
	AM	10.12	10.78
	NR	10.12	10.78
	NN	8.43	8.24
	ID	8.23	8.04
	Step-5 SMA	8.18	7.72

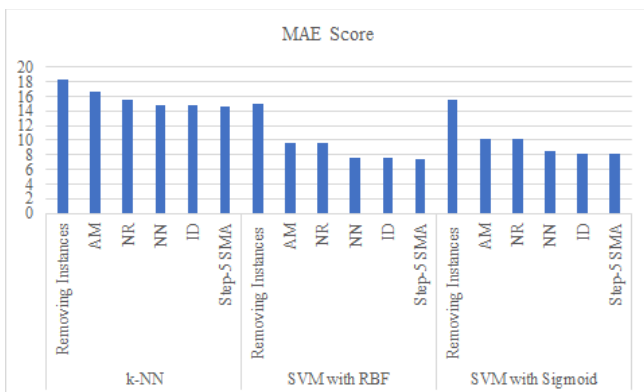


Fig. 2. Comparing results by MAE

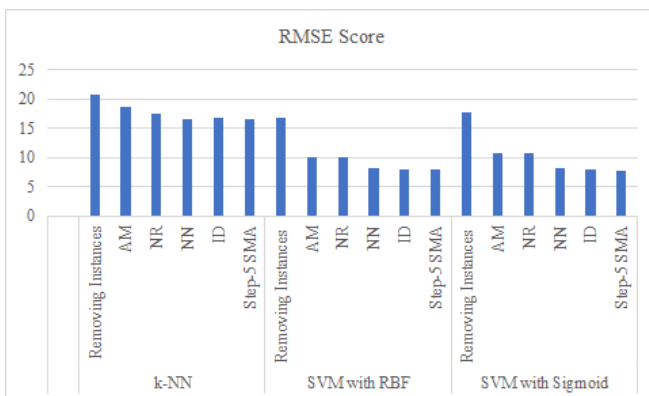


Fig. 3. Comparing results by RMSE

In addition, our proposed method (Step-5 SMA) outperformed the AM, NR, NN and ID methods based on

average MAE scores by 20.70%, 17.31%, 1.79% and 1.19%, respectively and outperformed AM, NR, NN and ID methods based on average RMSE scores by 22.37%, 19.29%, 2.52% and 1.74%, respectively. Including consideration of the transition period of the ENSO phenomenon for estimating missing values of rainfall data produced values closer to the actual values without greatly increasing computational time.

VII. CONCLUSION

Missing data impact rainfall forecasting accuracy and lead to biased estimates and invalid conclusions. Therefore, here, we compared methods of handling missing data values in historical monthly rainfall data. On average, 10% of the data were missing. Methods of handling the issue of missing data values in historical monthly rainfall data included instances of missing value removal and methods of estimating missing values as AM, NR, NN and ID. We also proposed a method of estimating missing values, called *step-5 SMA*. This was based on the simple moving average technique by taking the transition period of the ENSO phenomenon into consideration. The ENSO phenomenon changes every five years. Therefore, the number of total periods in this study should be five.

After using the enhanced training set to develop models of monthly rainfall forecasting by *k*-NN and SVR with RBF, and SVR with sigmoid, results showed that monthly rainfall forecasters developed by the training set that removed observations with missing values returned the lowest performance. Meanwhile, enhancing the quality of the training set using our *step-5 SMA* gave a better performance than the other missing value estimation methods. This indicated that our method was suitable as an optimal approach for estimating missing values, especially when using with this dataset.

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REFERENCES

- [1] M. Borga, E. N. Anagnostou, G. Blöschl, and J. Creutin. "Flash flood forecasting, warning and risk management: The HYDRATE project", *Environmental Science & Policy*, vol. 14, issue. 7, pp. 934-844, November 2011.
- [2] T. T. Luong, J. Pöschmann, R. Kronenberg, and C. Bernhofer, "Rainfall Threshold for Flash Flood Warning Based on Model Output of Soil Moisture: Case Study Wernersbach, Germany", *Water Journal*, vol. 13, issue. 8, pp. 1-17, April 2021.
- [3] D. Detchphol Chitwatkulsiri, H. Miyamoto, and S. Weesakul, "Development of a Simulation Model for Real-Time Urban Floods Warning: A Case Study at Sukhumvit Area, Bangkok, Thailand", *Water Journal*, vol. 13, issue. 11, pp. 1-17, May 2021.
- [4] B. Praveen, S. Talukdar, Shahfahad, S. Mahato, J. Mondal, P. Sharma, A. Reza, T. Islam, and A. Rahman, "Analyzing trend and forecasting of rainfall changes in India using non-parametrical and machine learning approaches", *Scientific Reports*, vol. 10, no. 10342, June 2020.

- [5] M. Raval, P. Sivashanmugam, V. Pham, H. Gohel, A. Kaushik, and Y. Wan, "Automated predictive analytics tool for rainfall forecasting", *Scientific Reports*, vol. 11, no. 17704, September 2021.
- [6] S. Kotsiantis, A. Kostoulas, S. Lykoudis, A. Argiriou, and K. Menagias, "Filling missing temperature values in weather data banks", 2006 2nd IET International Conference on Intelligent Environments, 5-6 July 2006.
- [7] F. Rafii and T. Kechadi, "Collection of historical weather data: issues with missing values", *Proceedings of the 4th International Conference on Smart City Applications*, pp. 1-8, October 2019.
- [8] A. Aieb, K. Madani, M. Scarpa, B. Bonaccorso, and K. Lefsih, "A new approach for processing climate missing databases applied to daily rainfall data in Soummam watershed, Algeria", *Heliyon*, vol. 5, issue 2, February 2019.
- [9] T.D. Pigott, "A Review of Methods for Missing Data", *Journal of Educational Research and Evaluation*, vol. 7, no. 4, pp. 353-383, 2001.
- [10] P. Royston, "Multiple imputation of missing values", *The Stata Journal*, vol. 4, no. 3, pp. 227-241, 2004.
- [11] K.L. Sainani, "Dealing With Missing Data", *PM&R Journal*, vol. 7, pp. 990-994, 2015.
- [12] N.M. Noor, "Roles of Imputation Methods for Filling the Missing Values: A Review", *Advances in Environmental Biology*, vol. 7, no. 12, pp. 3861-3869, 2013.
- [13] T.D. Baddoo, Z. Li, S.N. Odai, K.R.C. Boni, I.K. Nooni, and S.A. Andam-Akorful, "Comparison of Missing Data Infilling Mechanisms for Recovering a Real-World Single Station Streamflow Observation", *Int. J. Environ. Res. Public Health*, vol. 18, no. 2375, pp. 1-26, August 2021.
- [14] C.M. Salgado, C. Azevedo, H. Proença, and S.M. Vieira, "Missing Data", *Secondary Analysis of Electronic Health Records*, pp. 143-162, September 2016.
- [15] H. Kang, "The prevention and handling of the missing data", *Korean J Anesthesiol*, vol. 74, no. 5, pp. 402-406, May 2016.
- [16] T. Frisell, "Why Missing Data Is A Problem, and What You shouldn't Do To Solve It", *BMJ Journal*, vol. 75, issue 2, 2021.
- [17] W.N. Wan Ismail, "Estimation of rainfall and stream flow missing data for Terengganu, Malaysia by using interpolation technique methods", *Malaysian Journal of Fundamental and Applied Sciences*, vol. 13, no. 3, pp. 213-217, September 2017.
- [18] A. Shigute Boke, "Comparative Evaluation of Spatial Interpolation Methods for Estimation of Missing Meteorological Variables over Ethiopia", vol. 9, no. 8, July 2017.
- [19] T.R. Nkuna and J.O. Odiyo, "Filling of missing rainfall data in Luvuvhu River Catchment using artificial neural networks", *Physics and Chemistry of the Earth Parts A/B/C*, 2011.
- [20] W. Sanusi, W.A. Wan Zin, U. Mulbar, and M. Danial, "Comparison of the Methods to Estimate Missing Values in Monthly Precipitation Data", *International Journal on Advanced Science Engineering and Information Technology*, vol. 7, no. 6, December 2017.
- [21] M. Le, "Techniques of Filling Missing Values of Daily and Monthly Rain Fall Data: A Review", *SF Journal of Environmental and Earth Science*, vol. 2, no. 1036, pp. 1-5, 2020.
- [22] H. Hassani, M. Kalantari, and Z. Ghodsi, "Evaluating the Performance of Multiple Imputation Methods for Handling Missing Values in Time Series Data: A Study Focused on East Africa, Soil-Carbonate-Stable Isotope Data", *Stats*, vol. 2, no. 4, pp. 457-467, December 2019.
- [23] S. Yang, Z. Li, J.Y. Yu, X. Hu, W. Dong, and S. He, "El Niño–Southern Oscillation and its impact in the changing climate", *National Science Review*, vol. 5, issue 6, pp. 840–857, November 2018.
- [24] M. Huang, R. Lin, S. Huang, and T. Xing, "A novel approach for precipitation forecast via improved K-nearest neighbor algorithm", *Advanced Engineering Informatics*, vol. 33, pp. 89-95, August 2017.
- [25] H.H. Dawoodi, "Rainfall Prediction In North Maharashtra Region Using Support Vector Machine", *Turkish Journal of Computer and Mathematics Education*, vol. 12, no. 7, 2021.
- [26] N. Hasan, N.C. Nath, and R.I. Rasel, "A support vector regression model for forecasting rainfall", 2nd International Conference on Electrical Information and Communication Technologies (EICT), 10-12 December 2015.
- [27] T. Chai and R.R. Draxler, "Root mean square error (RMSE) or mean absolute error (MAE)?" *Geoscientific Model Development Discussions*, vol. 7, no. 1, pp. 1525–1534, January 2014.